

Class 01A: Introduction to Artificial Intelligence

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AI – the new trend ?

Started at a workshop at
Dartmouth College, 1956

John McCarthy



- **AlphaGo**'s (developed by Google DeepMind) historic victory in March 2016 against Go master Lee Sedol ... a landmark of the field of AI.
- AI is the new **UI**
- AI is making every interface both simple and smart.
- 1997 **Deep Blue** won Garry Kasparov



People interacting with technology through an AI
intermediator. e.g. mobile phone queries



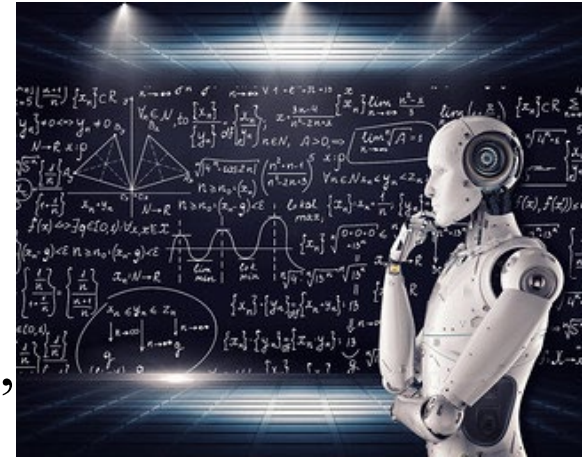
My short history of learning AI

University of Cambridge: expert systems

University of Waterloo: Neural Networks,

Prof. **Zadeh, Hinton**

The University of Hong Kong: intelligent control, mechatronics, intelligent transport systems.

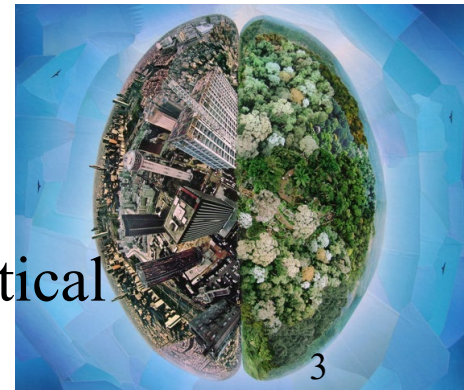


Breakthrough ... **ChatGPT** (Nov 2022)... mentors, sage

Hallucinations ... incorrect or misleading results of AI models

- ... different from wrong prediction
- ... news that contains incorrect facts
- ... biased views to manipulate public opinions

- The history of AI dates back to the 1950s
- **Alan Turing** proposed that machines could learn.
- Turing is widely considered to be the father of theoretical computer science. A code breaker ...





How many "r"s are there in the word strawberry?



GPT-3.5-Turbo

Poe

There are **two** "r"s in the word "strawberry."

Are you sure?



GPT-3.5-Turbo

Poe

Apologies for the confusion earlier. Let's count the number of "r"s in the word "strawberry" again:

There are **two** "r"s in the word "strawberry."

I think there should be three "r" in the word.

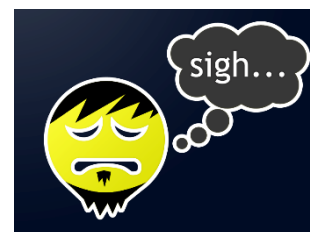
GPT 3.5

Aug 28, 2024

GPT-4.1-nano

Aug 11, 2025

chatgpt.hku.hk



I think there should be three "r" in the word.

 GPT-3.5-Turbo Poe

I apologize for the oversight. Let's count the number of "r"s in the word "strawberry" once more:

There are actually **three** "r"s in the word "strawberry." Thank you for pointing that out.

Please double check

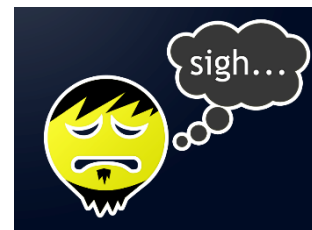
 GPT-3.5-Turbo Poe

Let's count the number of "r"s in the word "strawberry" one more time to be absolutely sure:

There are indeed **two** "r"s in the word "strawberry."

GPT 3.5
Aug 28, 2024

GPT-4.1-nano
Aug 11, 2025



Why AI ? What is AI ?

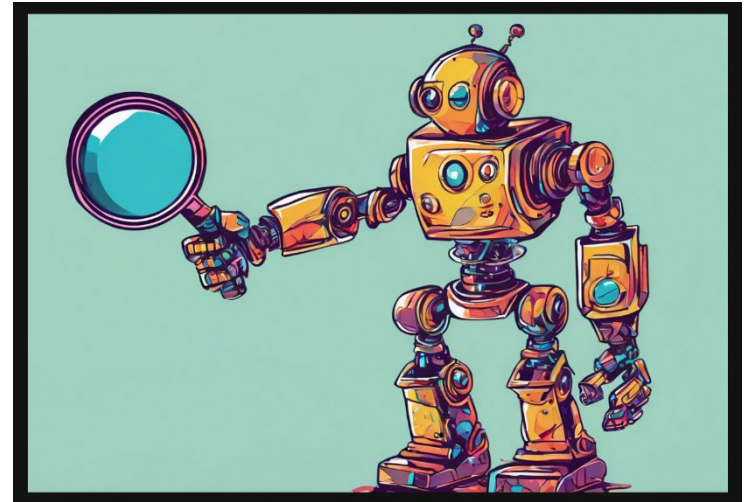
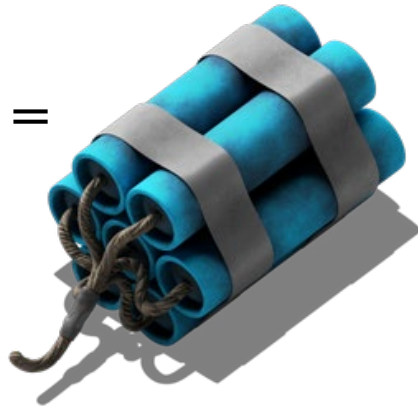
The scenario:

The CEO of a company goes to meet a technology consultant and asks:
“What kind of AI can I use?”

And the answer is, “You don’t need AI for this, and it will be
counterproductive for your aim[s].”

But the response is always, “But we won’t get any funding if our
product doesn’t have AI in it!”

Artificial Intelligence =
Natural Stupidity



AI is not magic. Use when the problem you’re solving is **intricate** and
that you **can’t possibly solve it the old way**.

The project leader **needs to know what it is first** ...

AI is no longer a competitive advantage – it is a necessity !

Fundamental Questions of AI

Can we build intelligent machine ?

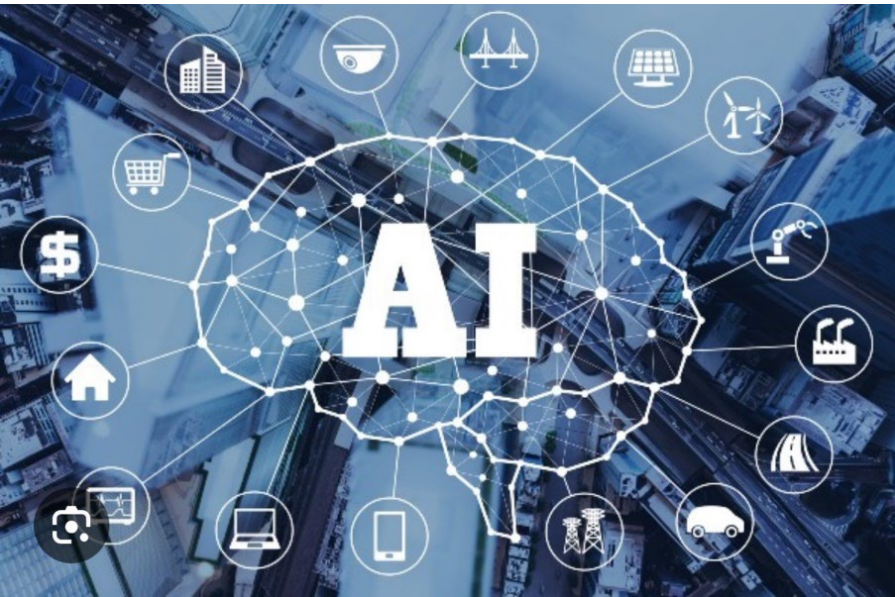
What is intelligence ?

Can machines think?

(*I think, there I am.*)



...





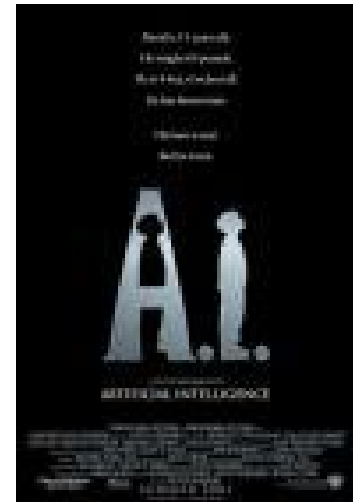
Discussion

- Can machines think?
- What are the attributes of intelligence ?
- How does a human mind work? Is it magic?
 - Can non-humans have minds?
 - Can machines replace human workers?
- Are intelligent machines good or bad for humans?
 - Would you trust one?



What is Intelligence?

- Intelligence is the ability to **understand** and **learn** things.
- Intelligence is the ability to **think** and understand instead of doing things by instinct or automatically. (*Essential English Dictionary*, Collins, London, 1990).
- Intelligence is the ability to **solve problems** and to **make decisions**.

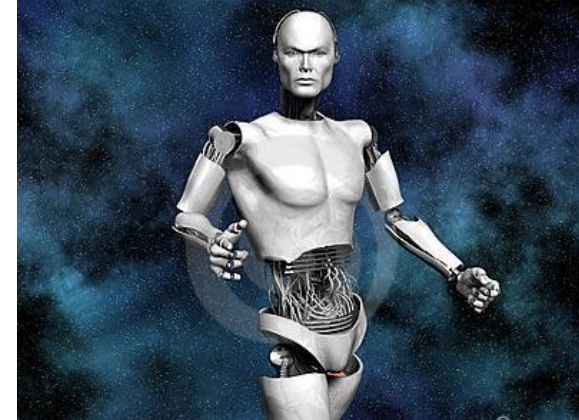


Are these tasks intelligent?

- a simple calculator
- a program to find a path through a network
- a system which 'sees' a component and controls a robot arm to build it into an assembly



Yes/No--- some say **AI is the study of how to make computers do things which at the moment humans are better!!!!**

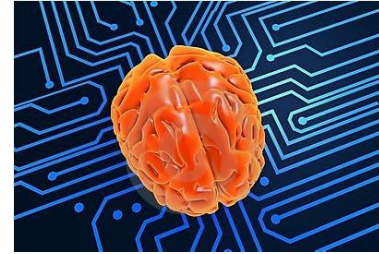


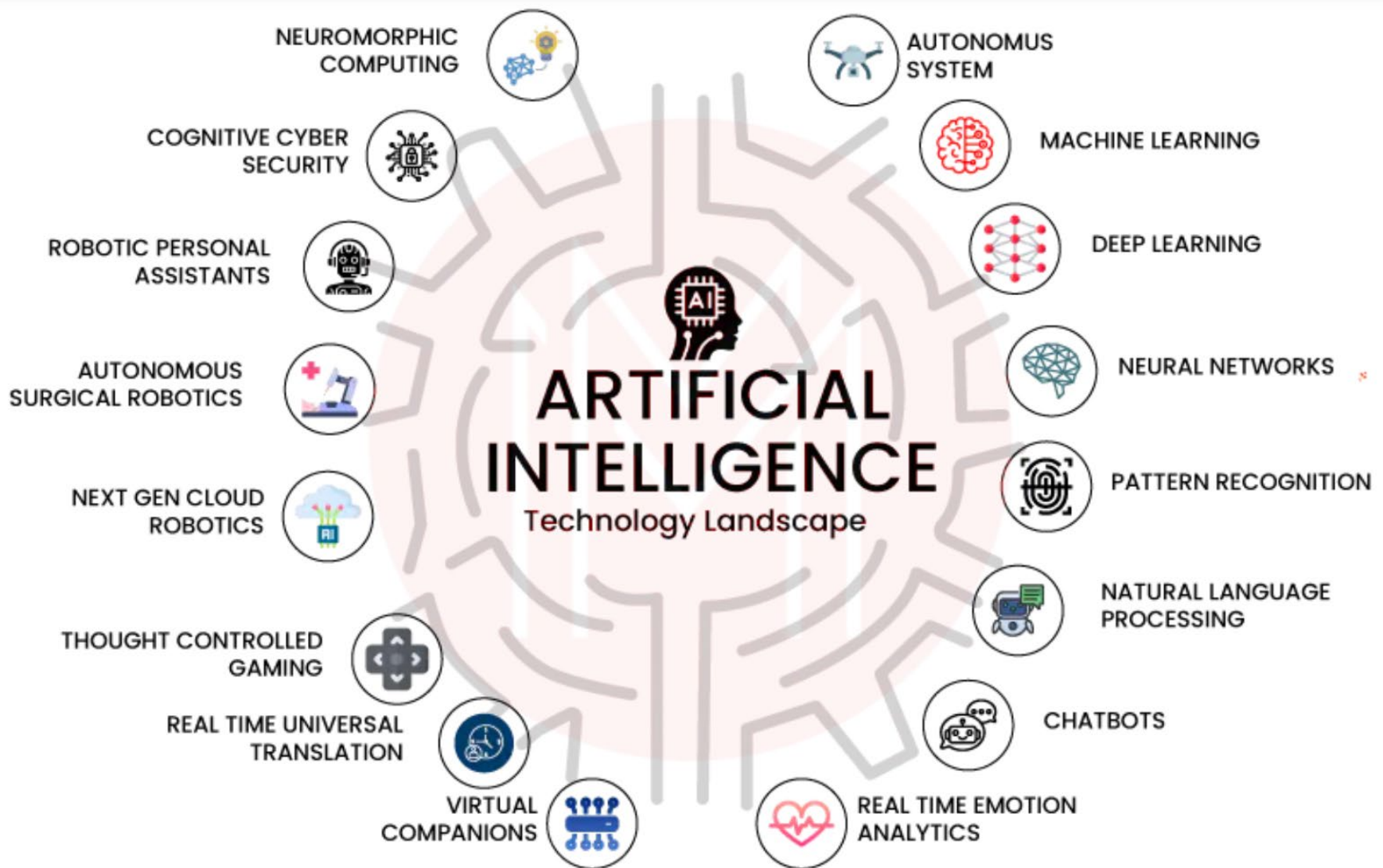
Yellow page enquiry: Dial 1083

Welcome to your telephone enquiry ...

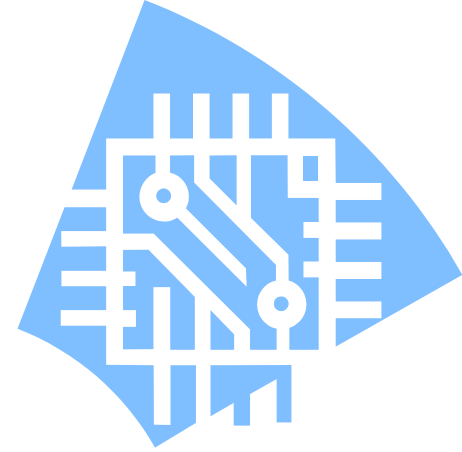
What is Artificial Intelligence?

- The application of computers to areas normally regarded as requiring human intelligence. Traditional problems (goals) are reasoning, knowledge representation, planning, learning, natural language processing.
- The goal of artificial intelligence (AI) as a science is to make machines do things that would require intelligence if done by humans.
- All areas of knowledge can help to develop intelligent systems.
- Artificial Cleverness: it has no true conscious understanding (Sir Roger Penrose, Fields Medalist Terence Tao)
- What is Machine Intelligence ?



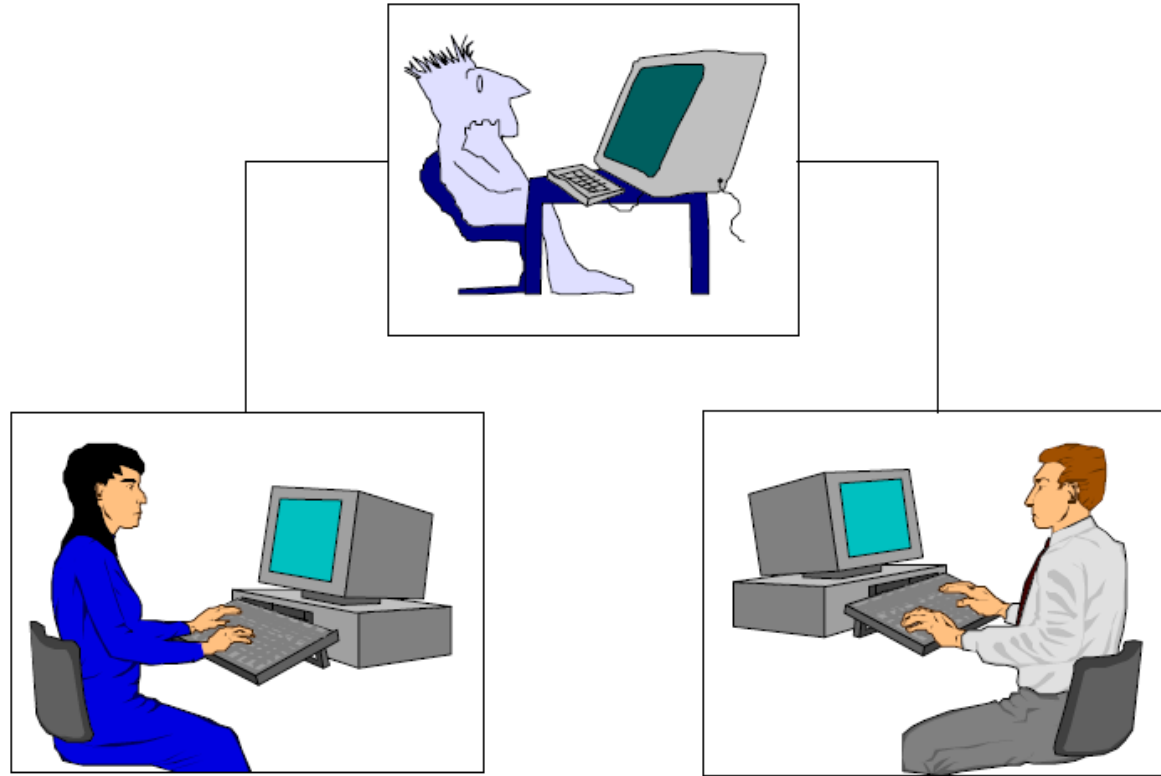


Turing Imitation Game



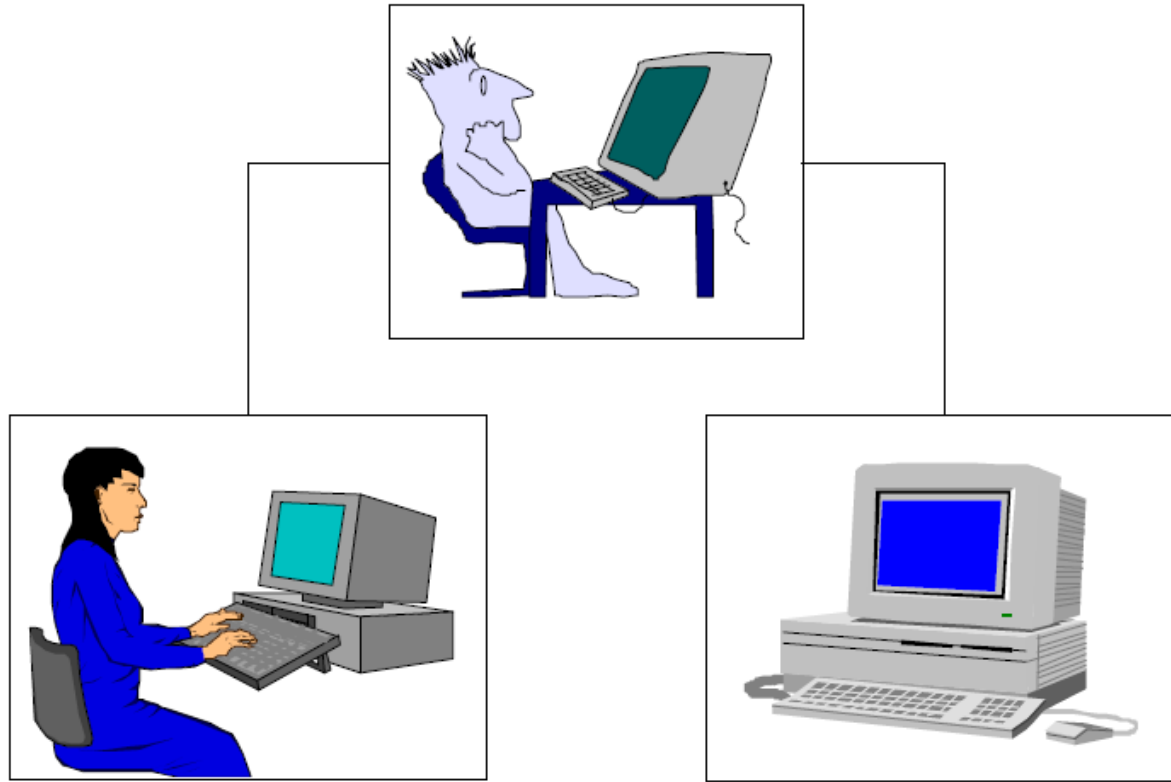
- The British mathematician Alan Turing invented a game in 1950 called the **Turing Imitation Game**.
- A test of a machine's ability to exhibit intelligent behaviour.
- A **benchmark** for intelligent program.
- The imitation game originally included two phases:

Turing Imitation Game – Phase 1



- In the first phase, the interrogator, a man and a woman are each placed in separate rooms. **The interrogator's objective is to work out who is the man and who is the woman by questioning them.**
- The man should attempt to deceive the interrogator that he is the woman, while the woman has to convince the interrogator that she is the woman.

Turing Imitation Game – Phase 2



- In the second phase of the game, **the man is replaced by a computer programmed to deceive the interrogator as the man did.** It would even be programmed to make mistakes and provide fuzzy answers in the way a human would.
- If the computer can fool the interrogator as often as the man did, we may say this computer has passed the intelligent behaviour test.

AI first, voice-mimicking software used in a major heist/robbery

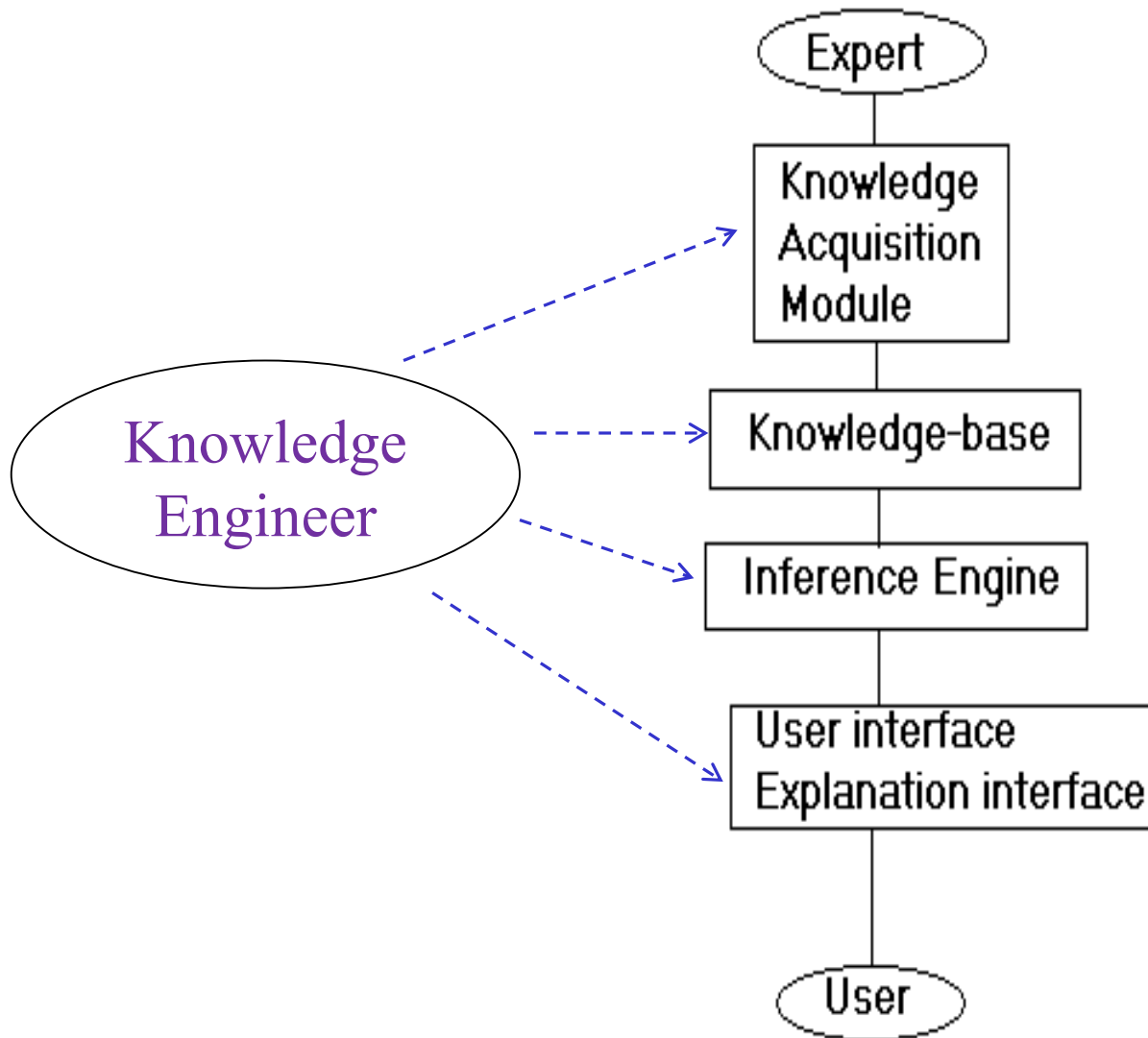
- A senior manager of a British energy company, believing his boss was on the phone, followed orders to wire US\$240,000 to an account in Hungary.
- The money transfer order was from a voice-mimicking software which imitated the company executive's speech.
- The voice was so lifelike that he felt he had no choice but to comply.
- Voice-synthesis software can copy the rhythms and intonations of a person's voice and be used to produce convincing speech.

Source: The Wall Street Journal, The Washington Post

Remarks by Professor Turing

- By maintaining communication between the human and the machine via terminals, the test gives us an **objective standard view on intelligence**.
- An “intelligent” program in some narrow area of expertise is evaluated by comparing its performance with the performance of a **human expert**.
- To build an intelligent computer system, we have to **capture, organise and use** human expert knowledge in some narrow area of expertise → **Expert Systems**. (available 24 hours)
- Expert systems consist of components: **Knowledge Base, Inference Engine, User Interface**. Expert advice, recommendation with explanations are expected.

A Typical “Expert” System



Characteristics of Expert Systems

Expert systems can be distinguished from conventional computer systems in that:

1. They *simulate human reasoning* about the problem domain, rather than simulating the domain itself.
2. They perform *reasoning* over *representations of human knowledge*, in addition to doing numerical calculations or data retrieval. They have corresponding distinct modules referred to as the *inference engine* and the *knowledge base*.
3. Problems tend to be solved using *heuristics* (rules of thumb) or *approximate methods* or *probabilistic methods* which, unlike algorithmic solutions, are not guaranteed to result in a correct or optimal solution.
4. They usually have to provide *explanations* and *justifications* of their solutions or recommendations in order to convince the user that their reasoning is correct.

Note that the term Intelligent Knowledge Based System (IKBS) is sometimes used as a synonym for Expert System.

CALL FOR PAPERS

The 11th International Conference on Artificial Intelligence (ICOAI 2024)

October 23-25, 2024 Dubai, UAE

- ICOAI 2024 allows academia to meet industry from the international community to exchange experiences, demonstrate their studies and further advance AI technologies.

Topics of Interest:

- Agent-based Systems
- Ant Colony Optimization
- Approximate Reasoning
- Artificial Immune Systems
- Artificial Intelligence in Modeling and Simulation
- Artificial Intelligence in Scheduling and Optimization
- Artificial Life
- Bioinformatics and Computational Biology
- Brain-Machine Interfaces
- Cognitive Systems and Applications
- Collective Intelligence
- Computer Vision
- Data Mining
- Differential Evolution
- Evolutionary Data Mining
- Evolutionary Design
- Evolutionary Scheduling
- Expert Systems
- Fuzzy Computing With Words
- Fuzzy Control and Intelligent Systems
- Fuzzy Decision Making and Decision Support Systems
- Fuzzy Logic and Fuzzy Set Theory

genetic algorithm

Topics of Interest:

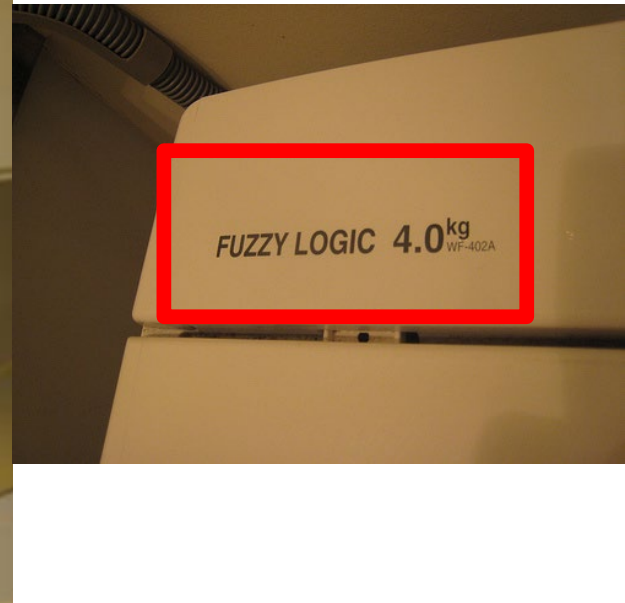
- Fuzzy Optimization and Design
- Fuzzy Pattern Recognition
- Fuzzy Systems for Robotics
- Game Theory
- Hardware Implementations
- Human–Computer Interaction
- Intelligent Database Systems
- Knowledge Engineering
- Machine Learning
- Modeling and Identification
- Molecular and Quantum Computing
- Multi-Agent Systems
- Natural Language Processing
- Neural Network Theory and Architectures
- Particle Swarm Optimization
- Robotics and Related Fields
- Rough Sets and Rough Data Analysis
- Speech Understanding
- Supervised and Unsupervised Learning
- Type-2 Fuzzy Logic
- Various Applications
- Web Intelligence

Intelligent devices/ machines around you (The application of AI)

- rice cooker







- **Japanese washing machine**



Brandt

A4 class
1000 rpm
1-7

Cotton				Synthetics				Delicates				Wool			
1 Intensive	90'	6 Normal	30'	10 Intensive	60'	11 Normal	50'	15 Normal	40'	18 Handwash	30'	19 Normal	30'	20 Cold	30'
2 Prewash	60'	7 FLASH40'		12 Normal	40'	13 Intensive	40'	16 Normal	30'	21 Cold	30'	22 Cold	30'	23 Cold	30'
3 Normal	60'	8 Cold	30'	14 Cold	30'										
4 Normal	40'	9 Mix	40'												
5 Eco	40'														

Start/Pause
Start/Pause
Start/Pause
Start/Pause

1000
850
700
400 rpm
550
Stop

Stop
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20

- **European washing machine**

Brandt



Cotton

- 1 Intensive 90°
- 2 Prewash 60°
- 3 Normal 60°
- 4 Normal 40°
- 5 Eco 40°

- 6 Normal 30°
- 7 FLASH 40°
- 8 Cold ❄️
- 9 Mix 40°



Synthetics

- 10 Intensive 60°
- 11 Normal 50°
- 12 Normal 40°
- 13 Intensive 40°
- 14 Cold ❄️



Delicates

- 15 Normal 40°
- 16 Normal 30°
- 17 Cold ❄️



Wool

- 18 Handwash 🧼
- 19 Normal 30°
- 20 Cold ❄️



Rinse



Spin



Drain

WFE1077A

AA class

1000 rpm

1 - 7 kg

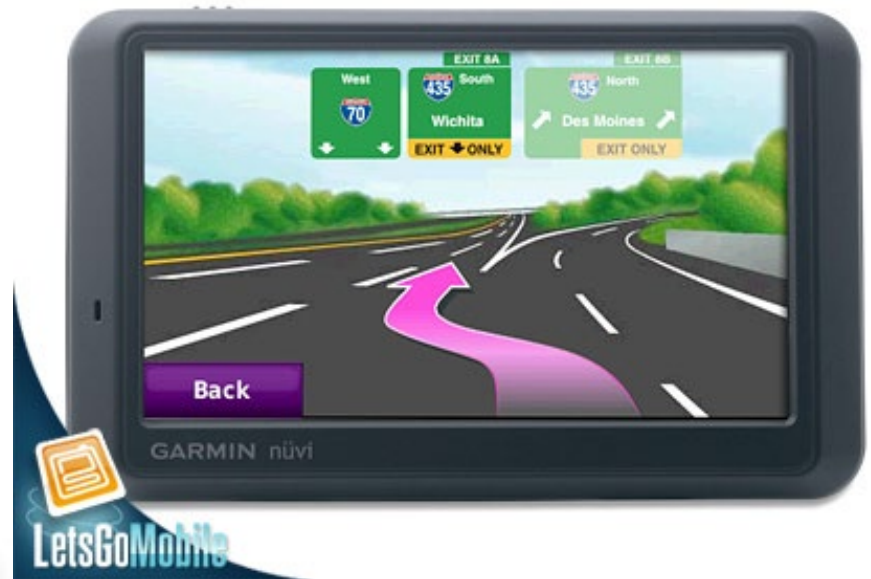
400 rpm
550
700



Stop



Car navigation system



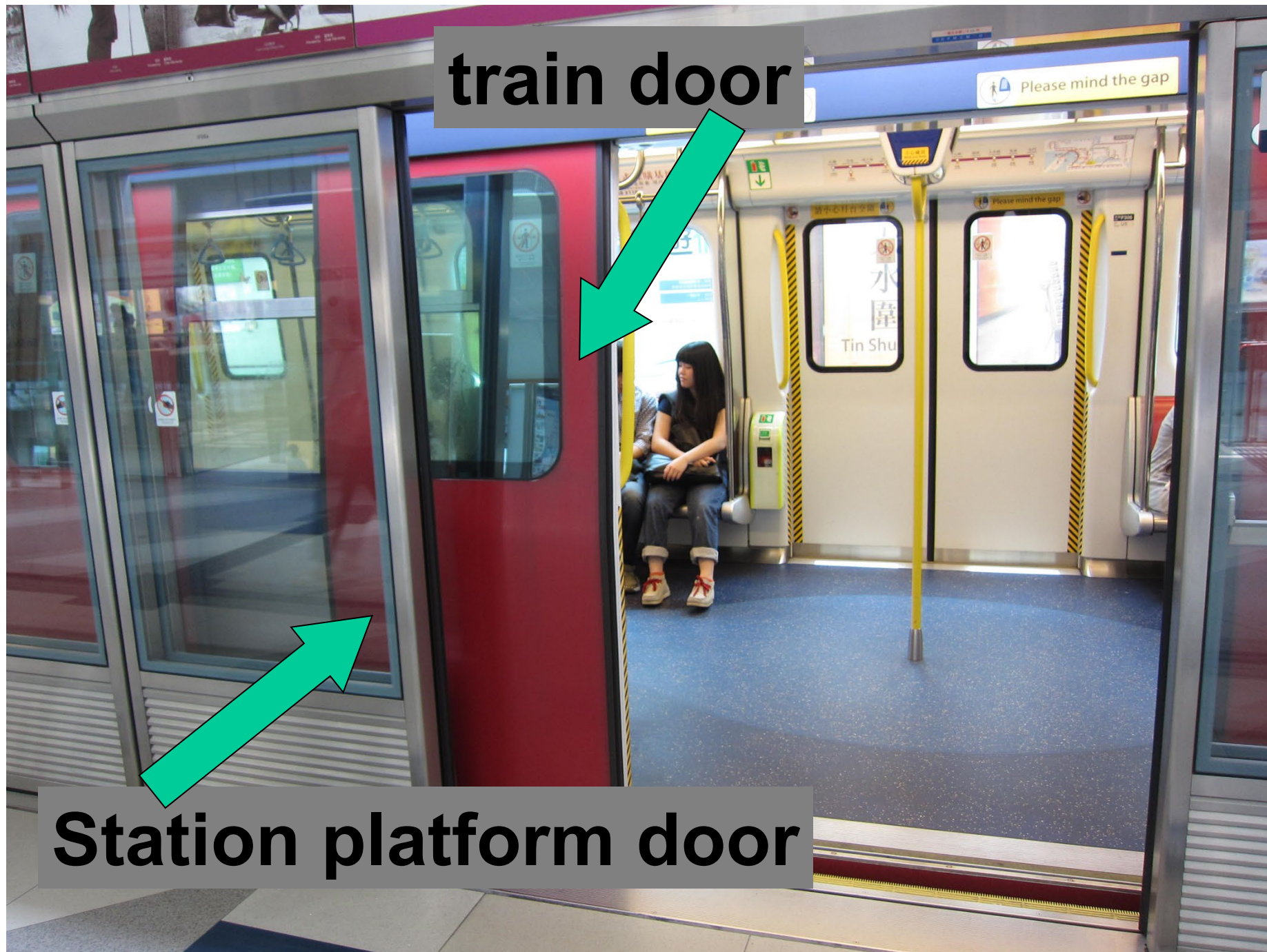
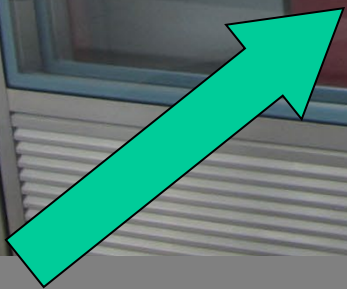




train door



Station platform door



Sendai 仙台市交通局電車

- The world's first train type to use **fuzzy logic** to control its speed, and this system (developed by **Hitachi**) accounts for the relative **smoothness of the starts and stops** when compared to other trains, and is 10% more energy efficient than human-controlled acceleration.
- The train can stop at **targeted positions.**



Approaches to AI (1)



- **Symbolic processing approaches**
 - **Physical symbol system hypothesis** [Newell & Simon]
 - “A physical symbol system has the necessary and sufficient means for general intelligence action”
 - Physical symbol system: **A machine (digital computer) that can manipulate symbolic data, rearrange lists of symbols, replace some symbols, and so on.**
 - Logical operations: McCarthy’s “advice-taker”
 - Represent “knowledge” about a problem domain by declarative sentences based on sentences in first-order logic
 - Logical reasoning to deduce consequences of knowledge
 - applied to declarative knowledge bases
 - **Prolog** is such a programming language; **LISP**

Approaches to AI (1)

– Top-down design method

- Knowledge level
 - Top level
 - The knowledge needed by the machine is specified
- Symbol level
 - Represent knowledge in symbolic structures (lists)
 - Specify operations on the structures, LISP
- Implementation level
 - Actually implement symbol-processing operations

Approaches to AI (1)

Rules for Knowledge Representation (in knowledge-based expert system)

RULES

- **IF** a person P complains of symptom S
AND drug D relieves symptom S
THEN person P should take drug D

FACTS

GLOBAL

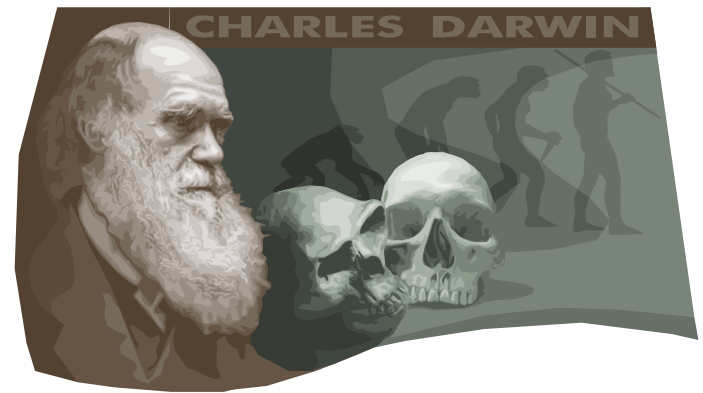
aspirin relieves headache

LOCAL

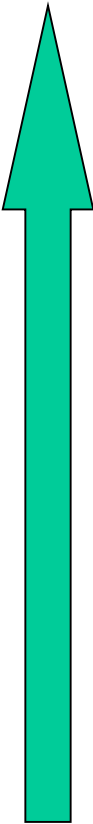
john complains of headache

A collection of rules and facts would form a KNOWLEDGE BASE.

Approaches to AI (2)



- **Subsymbolic processing approaches**
 - **Bottom-up style**
 - The concept of *signal* is appropriate at the lowest level
 - **Animate approach**
 - Human intelligence evolved only after a billion or more years of life on earth
 - Many of the same evolutionary steps need to make intelligence machines
 - **Training/Learning**
 - The complex behavior with the environment is learned during training

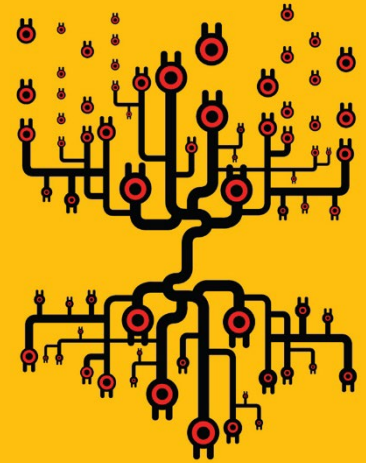


Approaches to AI (2)

- Well-known examples of machines coming from the **subsymbolic** school
 - Neural networks
 - Inspired by biological models
 - Ability to learn



- **Evolution** systems
 - Crossover, mutation, fitness



Brief History of AI



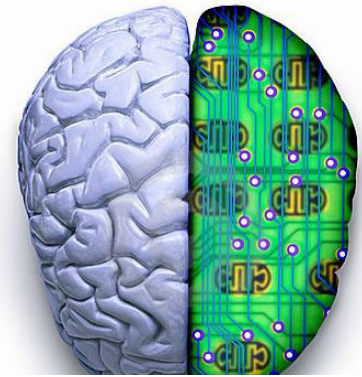
Symbolic AI

- 1943: Production rules
- 1956: “Artificial Intelligence”
- 1958: LISP AI language
- 1965: Resolution theorem proving
- 1970: PROLOG language
- 1971: STRIPS planner
- 1973: **MYCIN & DENDRAL expert systems**
- 1975: **Fuzzy Logic**
- 1982-92: **Fifth generation computer systems project**
- 1986: Society of mind
- 1994: Intelligent agents

Biological AI

- 1943: McCulloch-Pitt’s neurons
- 1959: Perceptron
- 1965: Cybernetics
- 1966: Simulated evolution
- 1966: Self-reproducing automata
- 1975: **Genetic algorithm**
- 1982: **Neural networks**
- 1986: Connectionism
- 1987: Artificial life
- 1992: **Evolution programming**
- 1994: DNA computing
- **Reinforcement learning**
- **Recurrent Neural Network**
- **Deep Learning (Convolutional NN)**

“Recent” History of AI



- **DEEP BLUE** (Sept 5, 1997)
 - Chess game playing program
 - IBM’s *Deep Blue* beats Kasparov!
- Human Intelligence
 - Ability to perceive/analyze a **visual** scene
 - Ability to understand and generate **language**
 - Winograd: Natural Language understanding system
 - LUNAR system: answer spoken English questions about rock samples collected from the moon
- IBM machine called **WATSON** wins a famous quiz contest (2011 summer) broadcasted in USA
- **AlphaGo** by Google DeepMind won Lee Sedol, awarded an honorary 9-dan (March 2016).

Central Issues in AI

- Knowledge representation
- Reasoning and control
- Learning (Knowledge acquisition)
- Handling uncertainty
- Forming plans/scheduling
- Vision (object recognition, motion and video analysis, scene understanding)
- Speech (recognition, synthesis, story comprehension)

Aim of the Course

- The aim is:
 - To demonstrate the **neural network** technique for **capturing** human knowledge.
 - To learn some architectures of neural networks.
 - To appreciate **deep learning** using CNNs
 - To use **fuzzy logic** to **mimic** the human ability to deal with incomplete or uncertain data.
 - Neural-Fuzzy method for learning
 - Genetic algorithm : intelligent search for solutions

Some early efforts on autonomous driving: DARPA Grand Challenge

- A prize competition for fully autonomous vehicles, funded by the **Defense Advanced Research Projects Agency**, the most prominent research organization of the United States Department of Defense.
- 2004 Grand Challenge (\$1 million)
- The first competition of the DARPA Grand Challenge was held on **March 13, 2004** in the Mojave Desert region of the United States, along a 150-mile (240 km) route.
- None of the robot vehicles finished the route. Carnegie Mellon University's Red Team and car Sandstorm traveled the farthest distance, completing 11.78 km (7.36 miles) of the course. The red team won that year.

DARPA Grand Challenge

- 2005 Grand Challenge (\$2 million)
- Five vehicles successfully completed the race.
- Winner: Stanford's Stanley (time used was 6:54)



DARPA Grand Challenge

- **2007 Urban Challenge** (\$2 million)
- Site: Southern California Logistics Airport, in Victorville, California
- The course involved a 96 km (60-miles) urban area course, to be completed in less than 6 hours.
- The \$2 million winner was Tartan Racing, a collaborative effort by **Carnegie Mellon University** and **General Motors Corporation**, with their vehicle "Boss", a Chevy Tahoe. Speed was averaged approximately 14 mph (22.53 km/h) throughout the course.
- Carnegie Mellon University has spent USD 40 million in the project !!

Tartan Racing is united to catalyze a technical, cultural and industrial revolution for a new class of robotics to advance the state-of-the-art in driver safety.



Winner of DARPA 2007 Grand Challenge



Red Whittaker serves as team lead for the Urban Challenge 2007. Premiere researchers from the Robotics Institute faculty are united as members of Tartan Racing. Chris Urmson, the team's technology leader, directs a team of world-class researchers, who have united to succeed in this competition. These include Tony Stentz, Martial Hebert, John Dolan, Bryan Salesky, Raj Rajkumar, Dave Ferguson and Paul Rybski from Carnegie Mellon and Varsha Sedakar and Bakhtiar Litkouhi from General Motors.

Other partners include Caterpillar, Continental AG, Intel, Google, Applanix, TeleAtlas, Vector, Ibeo, Mobileye, CarSim, CleanPower Resources, MA/COM, NetApp, Vector CANtech and Hewlett Packard.

2012 DARPA Grand Challenge (April 10, 2012)



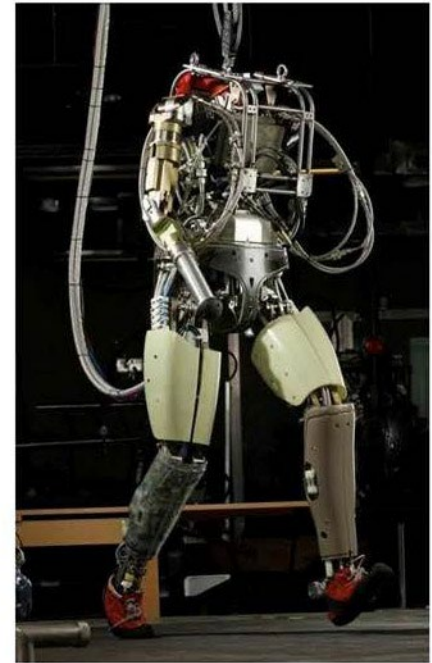
- DARPA Robotics Challenge, for disaster response operations.
- Biped design: used in rough terrain and for industrial disasters
- The robots can use human tools (hand tools or vehicles)

DARPA Robotics Challenge

- The robot will be required to **drive** a vehicle at the site
- **Travel** across rubble to proceed to a building
- Remove debris blocking the entry
- **Open** a door and enter into the building
- **Climb a ladder, and Traverse** a 100 meter rubble-strewn hallway
- **Use a tool to break** through a concrete panel
- **Locate** a leaking pipe and seal it by closing off a nearby valve
- **Replace** a faulty pump to resume normal operations -- all semi-autonomously with just “supervisory teleoperation”

DARPA Robotics Challenge

The DARPA Robotics Challenge is a response to the Fukushima Daiichi nuclear accident, where robots were ill-equipped to assess the disaster, inspect the facility, and fix problems. Its goal is to develop ground robots that can work in dangerous, degraded, human-engineered environments.



HKU Advanced Robotics Initiatives (July 2013)

Goals:

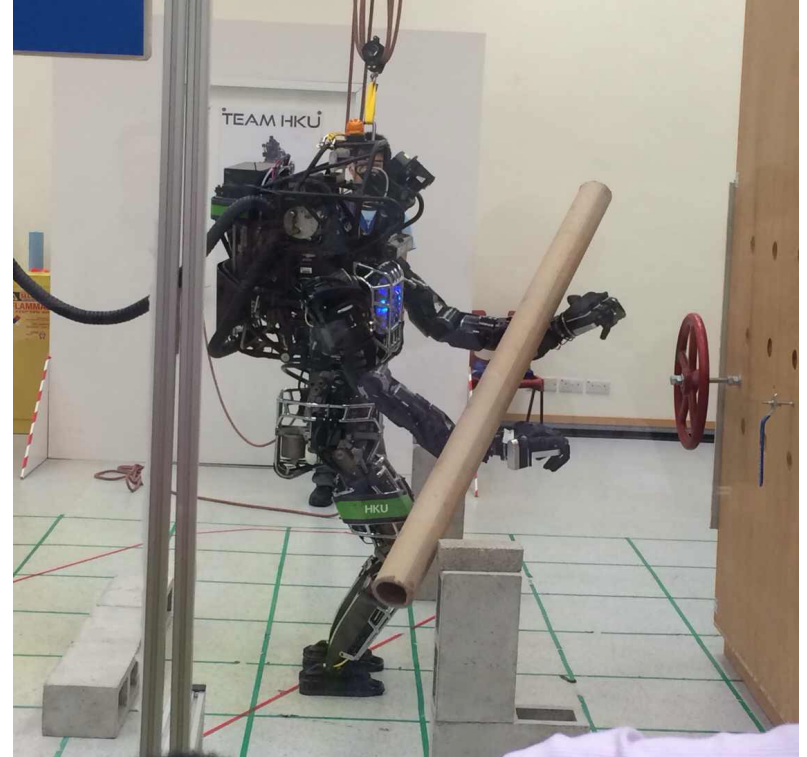
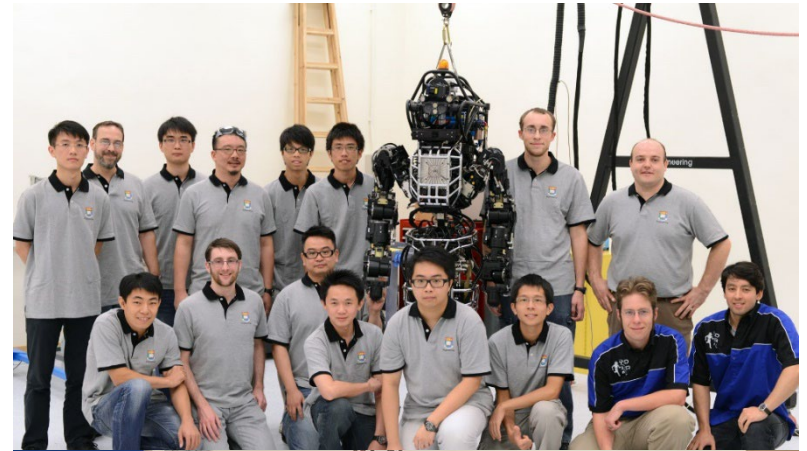
- Become world-class player and Asian leader in the area of robotics
- Development of ground-based robotic capabilities to execute complex tasks

HKU Advanced Robotics Initiatives

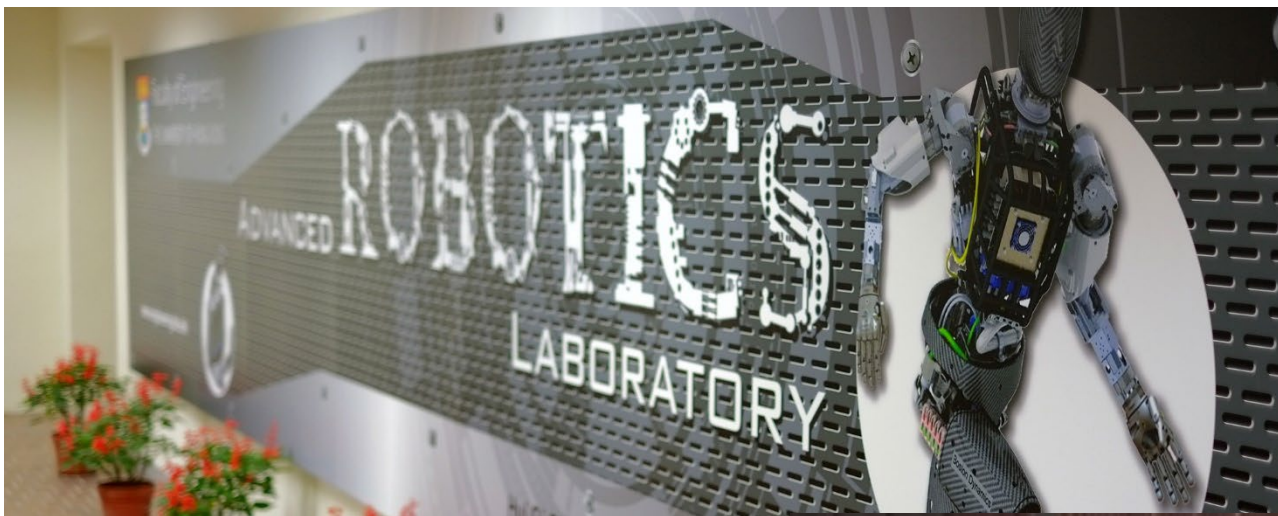
- <http://engg.hku.hk/home/robotics/news.htm> (see videos)
- <http://engg.hku.hk/home/robotics/>

Benefits:

- Spin-off technologies will be great assets to Hong Kong
- Smart Advanced Manufacturing
- Health care (i.e. smart wheelchairs)
- Disaster relief (search and rescue, firefighting)
- Hazardous maintenance and operations (utilities, airport, service tunnels, deep water, steep slope work)



HKU Advanced Robotics Initiatives



<https://www.youtube.com/watch?v=fn3KWM1kuAw>

← → ↺ youtube.com/watch?v=LikxFZZO2sk

Do you love me ? (2020)

☰ YouTube HK

<https://www.youtube.com/watch?v=tF4DML7FIWk>

Partners (2021)



What is the latest challenge ?

During one week in Sept 2021, some of the most sophisticated robots in the world were gathered deep below Louisville, Kentucky, USA, in a network of limestone caverns, caves and tunnels.

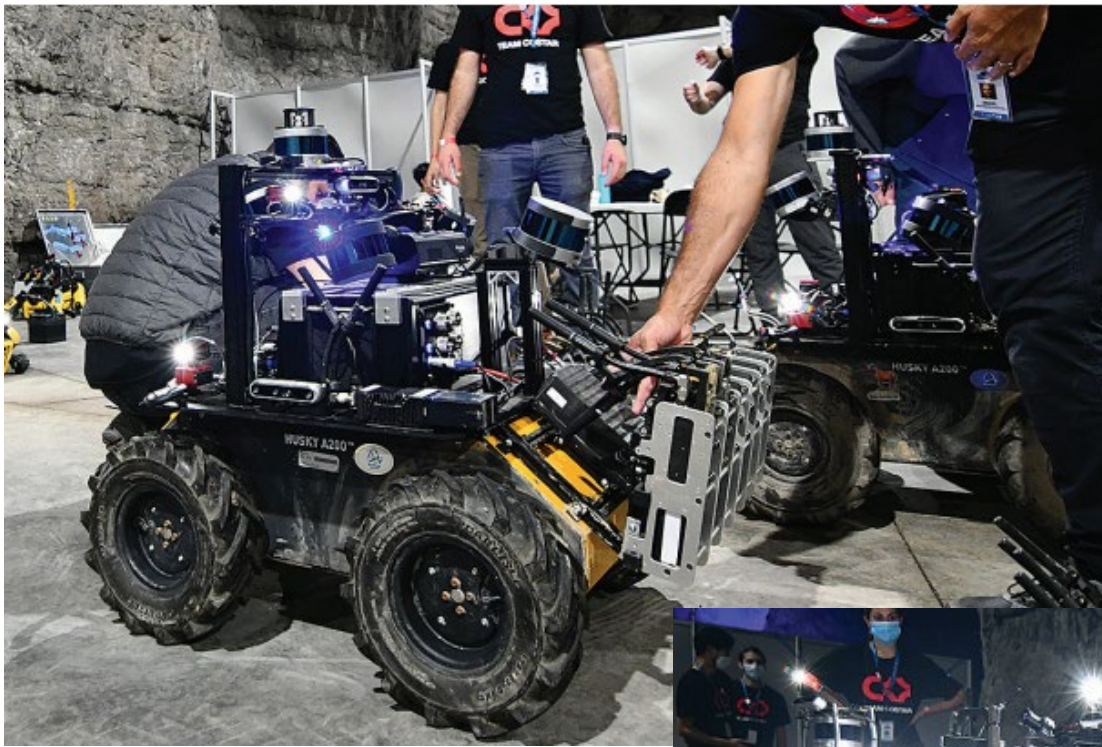
The place was dark, cold, dusty and humid.

DARPA's Subterranean Challenge

IEEE Spectrum

**Underground
With DARPA's
SubT Challenge**

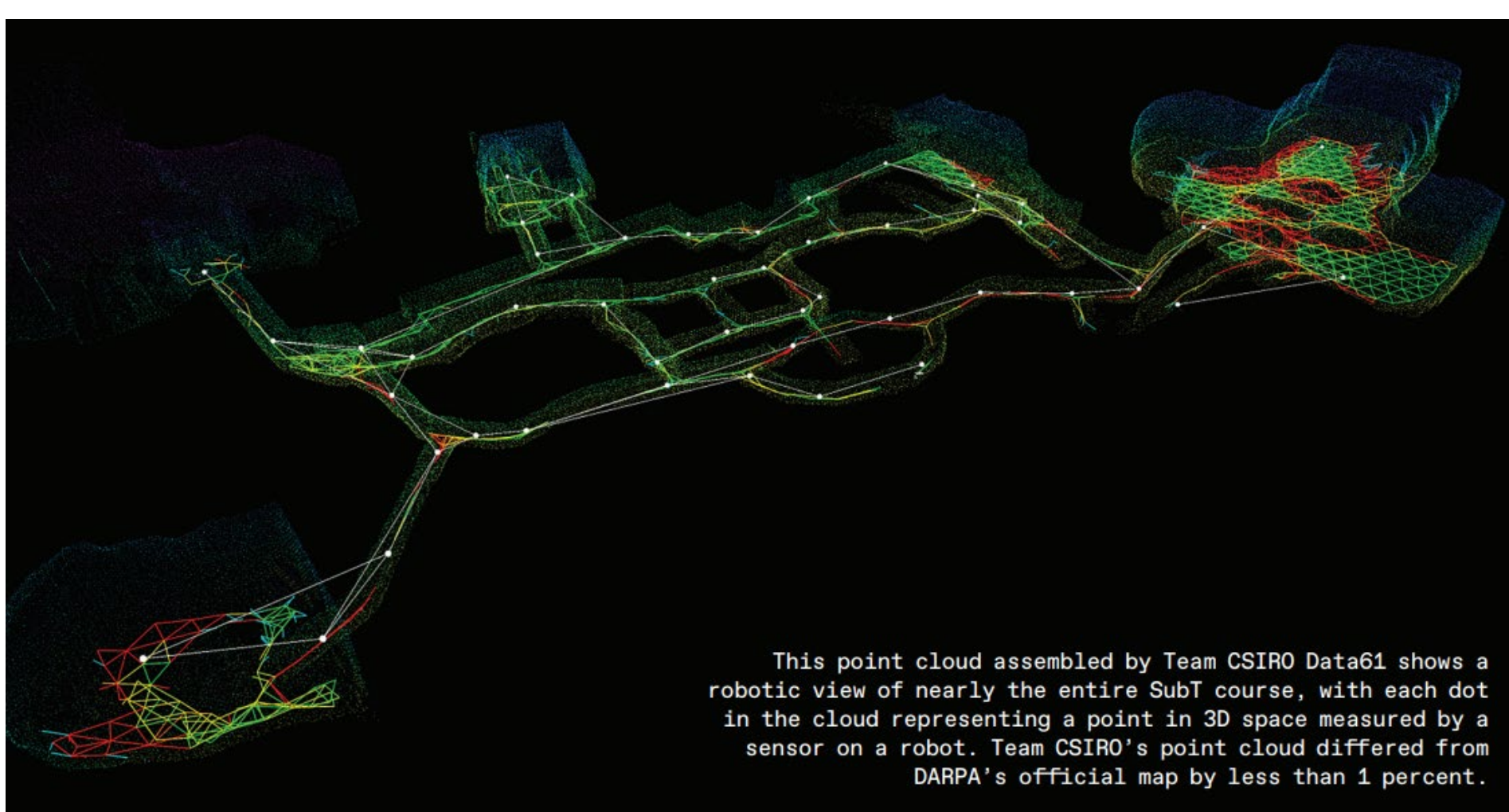




Team CoSTAR's
Husky wheeled
robots

Team CoSTAR's
Spot robots
(quadrupedal robots)





SubT course map gathered from a robot sensor

- Communications is a big problem
- Autonomy: only one person can interact remotely with the team's robots
- To find the 40 artifacts in the tunnel/caves.



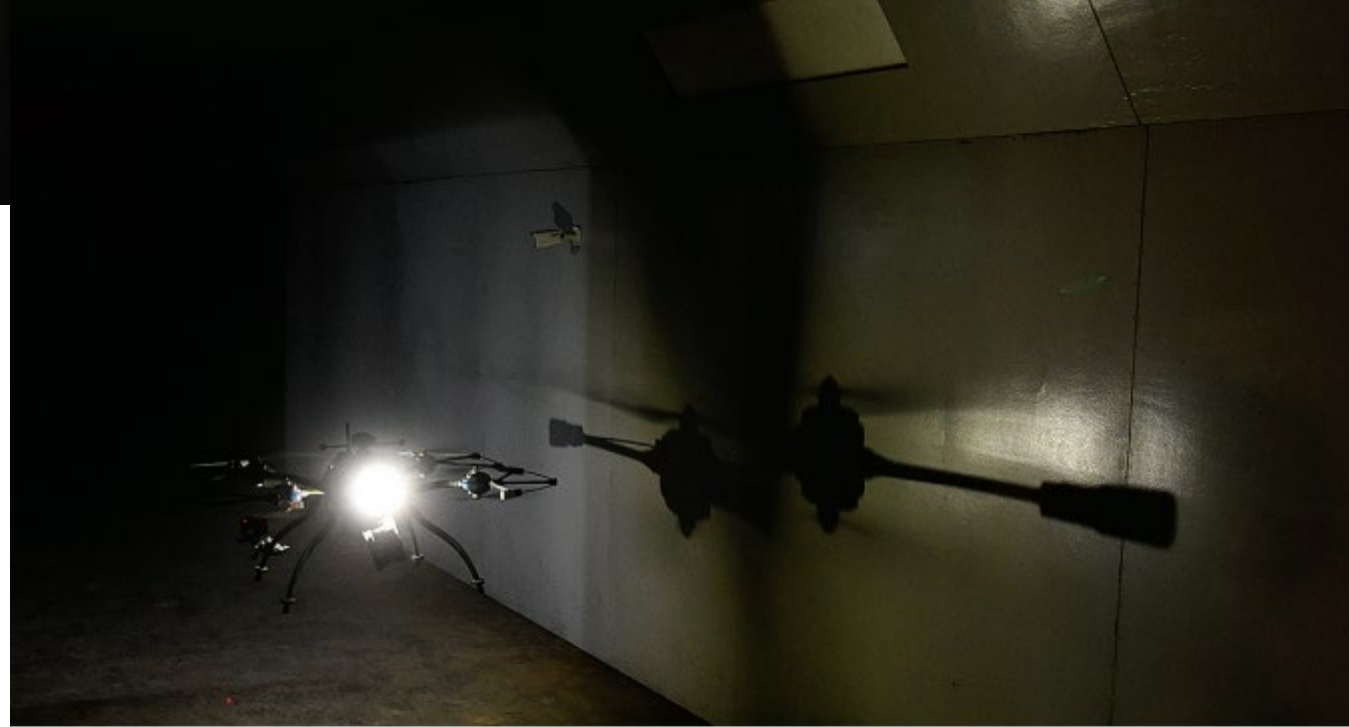
Eight teams in the
SubT Final.

Team had one hour
to find the 40
artifacts.

In the end, two
teams had each
found 23 artifacts.

Team Cerberus was
46 seconds faster
than Team CSIRO.





Drones can get disoriented in a blank hallway with no unique features

While most of the course was designed to look as much like real underground environments as possible, DARPA also included sections that posed very robot-specific challenges. Robots had the potential to get disoriented in this blank white hallway (part of the urban section of the course) if they couldn't identify unique features to differentiate one part of the hallway from another.

AI in Healthcare and Business

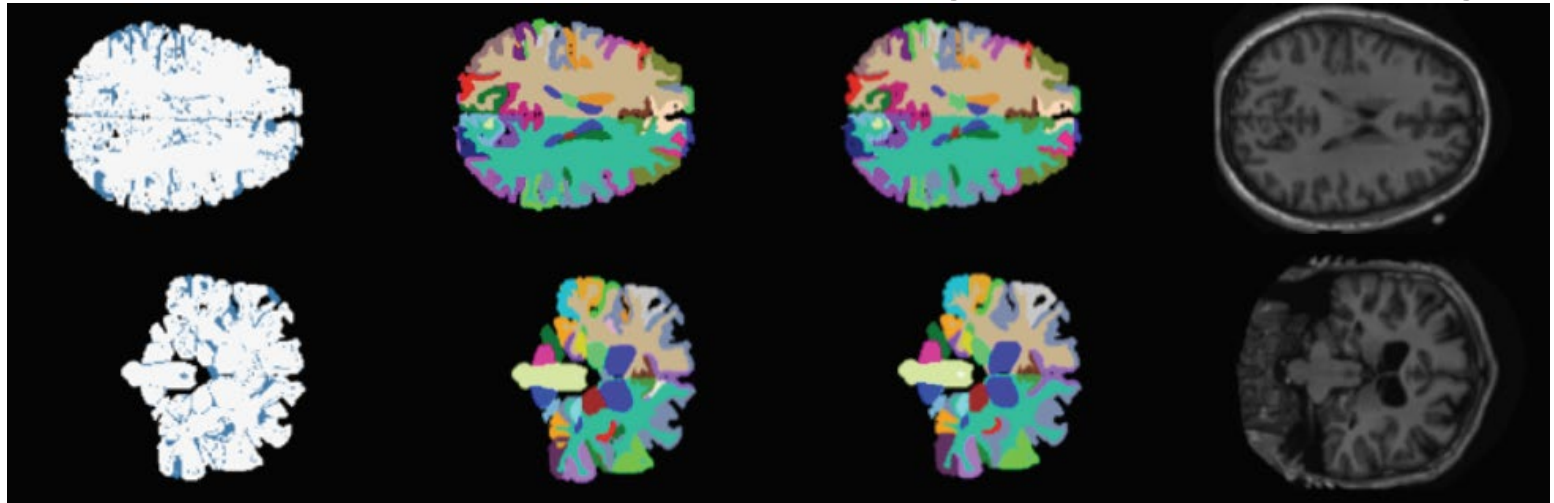
- MRI image segmentation

difference

CNN result

manual segmentation

MRI image



- AI for business executives: “This workshop is for executives and professionals who want to learn about the capabilities of **Artificial Intelligence, Machine Learning and Deep Learning** that are coming to pervade the world and workplace and are finding acceptance in almost all the industries and **sectors to solve many practical problems** of cost management, sales growth and technological innovation.” e.g. **credit card application approval, mortgage approval**
- “According to Gartner, by 2020, AI is projected to create 2.3 million new **jobs** worldwide while eliminating 1.8million traditional jobs.”

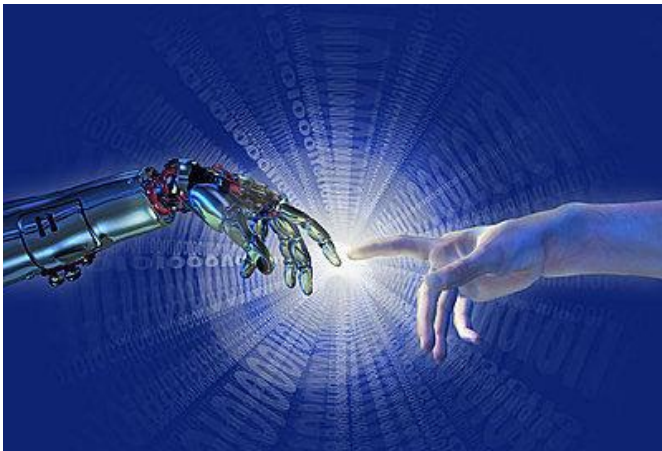
AI ... What is new ?

Artificial Intelligence(AI)

- Because of new computing technologies, **machine learning** today is not like machines learning of the past. It was born from pattern recognition and the theory that **computers can learn without being programmed to perform specific tasks**; researchers interested in artificial intelligence wanted to see if computers could learn from data.
- The iterative aspect of machine learning is important because as models are exposed to **new data** they are able to independently **adapt**.
- **Machine learning** has gained fresh momentum.
- **Deep Learning:** Convolutional Neural Networks
- From **GEN AI** towards **PHYICAL AI** ... FSD, Robots

Conclusions

- AI → Machine Learning
- Deep Learning in all areas
- Automation of Decision Making
- Makes life and the world more interesting (or dangerous) !
- Black Mirror
(MetalHead)



The Danger



- Ethnical Issues
- AI in Military ...
- AE (Artificial Emotions)
 - AI to mimic human motions ...
 - **“I am sorry. I understand the sadness with you.”**
 - AE can mimic/express both positive and negative motions ...
 - AE can manipulate human emotions too. AI for counselling.
 - AI sales assistant to make you buy more online products.

A Brave New Dawn ?